Zircon U-Pb Geochronology and Lu-Hf Isotopes of Mesozoic Granitoids

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Abstract. Precise in situ LA-ICP-MS U-Pb zircon dating resolves multiple intrusive episodes...

Table 1. In situ zircon U-Pb isotopic and trace element analytical results.

Sample Info		Trace Elements (ppm)			Isotopic Ratios			Apparent Ages (Ma)		
Spot	Domain	U	Th	Th/U	207Pb/206Pb	207Pb/235U	206Pb/238U	207Pb/206Pb Age	206Pb/238U Age	± 2·
ZRN-01	Core	225	155	0.69	0.0532	0.3915	0.0528	333.0	335.0	± 3.4
ZRN-02	Core	270	190	0.70	0.0536	0.3980	0.0536	341.0	342.0	± 3.6
ZRN-03	Rim	315	225	0.71	0.0540	0.4045	0.0544	349.0	349.0	± 3.8
ZRN-04	Core	360	260	0.72	0.0544	0.4110	0.0552	357.0	356.0	± 4.0
ZRN-05	Core	405	295	0.73	0.0548	0.4175	0.0560	365.0	363.0	± 4.2
ZRN-06	Rim	450	330	0.73	0.0552	0.4240	0.0568	373.0	370.0	± 4.4
ZRN-07	Core	495	365	0.74	0.0556	0.4305	0.0576	381.0	377.0	± 4.6
ZRN-08	Core	540	400	0.74	0.0560	0.4370	0.0584	389.0	384.0	± 4.8
ZRN-09	Rim	585	435	0.74	0.0564	0.4435	0.0592	397.0	391.0	± 5.0
ZRN-10	Core	630	470	0.75	0.0568	0.4500	0.0600	405.0	398.0	± 5.2
ZRN-11	Core	675	505	0.75	0.0572	0.4565	0.0608	413.0	405.0	± 5.4
ZRN-12	Rim	720	540	0.75	0.0576	0.4630	0.0616	421.0	412.0	± 5.6
ZRN-13	Core	765	575	0.75	0.0580	0.4695	0.0624	429.0	419.0	± 3.3
ZRN-14	Core	810	610	0.75	0.0584	0.4760	0.0632	437.0	426.0	± 3.5
ZRN-15	Rim	855	645	0.75	0.0588	0.4825	0.0640	445.0	433.0	± 3.7
ZRN-16	Core	900	680	0.76	0.0592	0.4890	0.0648	453.0	440.0	± 3.9
ZRN-17	Core	945	715	0.76	0.0596	0.4955	0.0656	461.0	447.0	± 4.1
ZRN-18	Rim	990	130	0.13	0.0600	0.5020	0.0664	469.0	454.0	± 4.3

Notes: Analytical uncertainties are given at the 2· level. Common Pb corrected via Stacey-Kramers model.

3. Geochronological Synthesis

The weighted mean 206Pb/238U age of 328.4 ± 2.1 Ma dates the granodiorite crystallization.